Track 1 Problem StatementRebuilding the Course Scheduling System

> **Investigate how the school currently schedules its classes, and redesign a scheduling system that makes the school's timetables even better while satisfying all of its constraints.**

1. The Problem

Before every semester, the school has to produce a timetable that takes care of teachers, students, classrooms and courses all at once. This timetable affects the daily study and work of teachers and students, so having a high-quality timetable is important.

We want you to consider the constraints in the school's timetable and optimize the timetable. By this, you should design a system and be able to explain why it is better than what we currently have. An example of a constraint is class hours: you may change Math class with English class, but the hours spent on math class in a semester should not change. The result should be a system that automatically schedules timetables.

There is no standard answer. How the current system works, which constraints must be satisfied, which aspects are worth improving: all of these are for you to research, judge and trade off, and you should explain your reasoning in your submission. During the presentation, the judges will ask why you made these decisions.

2. Evaluating Your Results

This problem cares more about which aspects you think can be improved and how you improve them. Your submission should answer:

- What problems in the current timetable system did you identify?
- Which aspects did you choose to optimize (for example, minimizing student movement between classrooms)? How did you optimize them?
- Who do these aspects affect, and how much? Why do they really matter?
- Compared with the current timetable, how much does your solution improve things? How do you measure it?

- How do you confirm that your timetable satisfies every constraint it must satisfy?
- Can your system still cope when circumstances change?

How to present your results most convincingly is also up to you.

3. Opportunities Behind This Problem

The school may apply outstanding solutions to its actual scheduling. This is not guaranteed and depends mainly on the quality and thoughtfulness of the solution.